



GROWTERN
Academy

45-DAYS SUMMER INTERNSHIP ON AI-POWERED **JAVA DEVELOPER**

MODULE : 01 JAVA FUNDAMENTALS

- Introduction to Java & setup (JDK, IDE)
 - Variables, data types, operators
 - Input/output (Scanner)
 - Conditional statements (if-else, switch) & Loops (for, while, do-while)
 - Mini Task: Basic Java programs
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MODULE : 02 OOP CONCEPTS

- Classes & objects
 - Constructors & methods
 - Inheritance & Polymorphism
 - Encapsulation & abstraction
 - Practice programs
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MODULE : 03 ADVANCED JAVA CONCEPTS

- Arrays & Strings
- Exception handling
- File handling (I/O streams)
- Collections Framework (List, Set, Map)
- Mini Project: Student Management System





MODULE : 04 **DATABASE & APPLICATION DEVELOPMENT**

- Introduction to JDBC
 - Database connectivity (MySQL)
 - CRUD operations
 - Basic GUI (Swing/JavaFX)
 - Application structure
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MODULE : 05 **PROJECTS & DEPLOYMENT**

- **Real-world projects:**
 - **Student Management system**
 - **ATM Banking Management System**
 - **Online Exam System**
 - Introduction to Git & GitHub
 - Project presentation & evaluation
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AI TRAINING MODULE : **GENERATIVE AI & AI TOOLS FOR DEVELOPERS**

- Introduction to Generative AI and its role in modern development
 - Introduction to AI tools: ChatGPT, Google Gemini, Claude, GitHub Copilot, Perplexity, ElevenLabs, SuperBot
 - Prompt engineering and effective AI usage
 - Code generation using AI tools for frontend and backend development
 - Building a mini project using AI-assisted workflow
 - Understanding limitations, ethics, and best practices of AI
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OUTCOME

Students will gain strong Java programming fundamentals, understand object-oriented concepts, and develop real-world applications with hands-on project experience.

